

Segment Anyword: Mask Prompt Inversion for Open-Set Grounded Segmentation

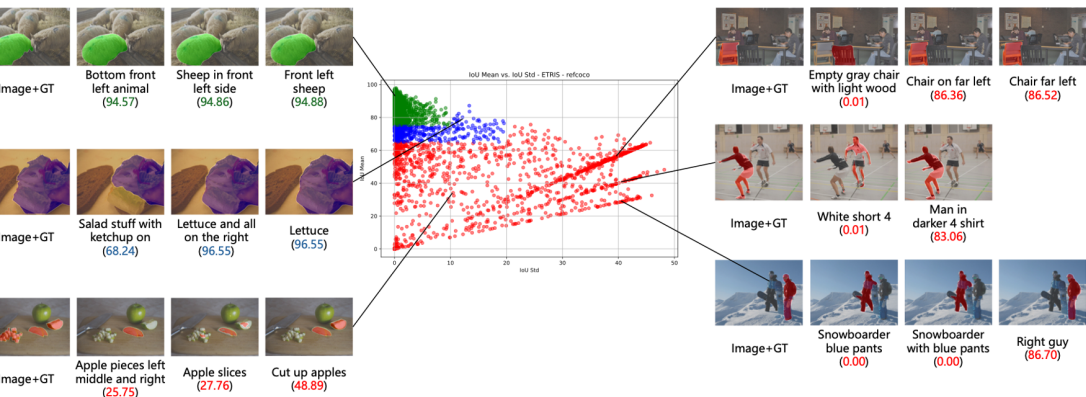
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Challenge

- Different sentences, but with similar description of an image.
- Open-Set (free-form image reference expressions).
- Heavy computational costs on training / fine-tuning VLM/MLLM

Setup: Reference Image Segmentation

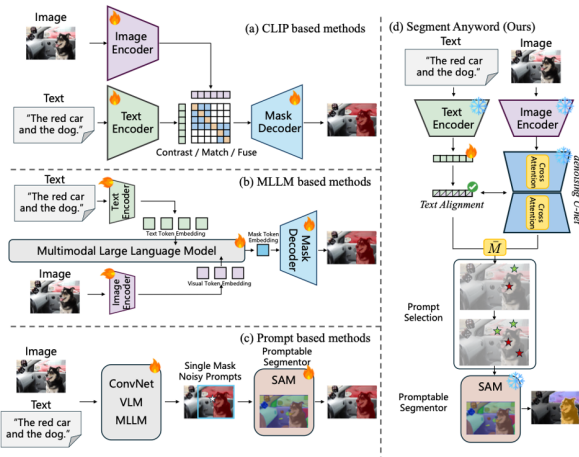


Observation

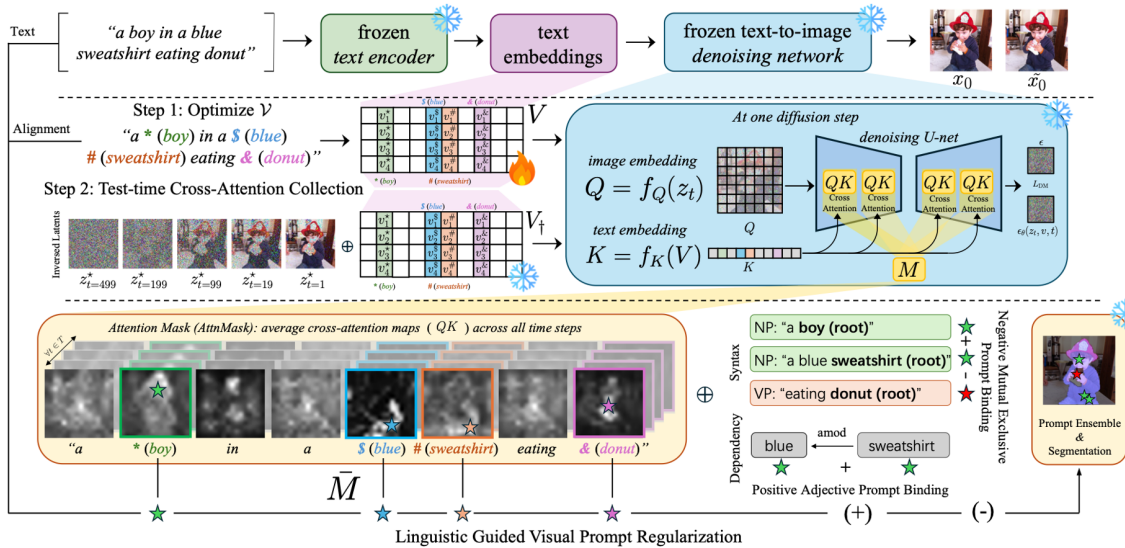
- Strong VLM/MLLM **NOT** stable (sensitive) to expression variation.
- Confounded from expression variation and image complexity.
- A lightweight, test-time adaptation method is desired.

Contribution

- Segment Anyword: A training-free test-time prompt-learning method encodes token-level visual concept.
- Linguistic guided regularization techniques for prompt binding and clustering.
- Superior performance among training free methods across several metrics.



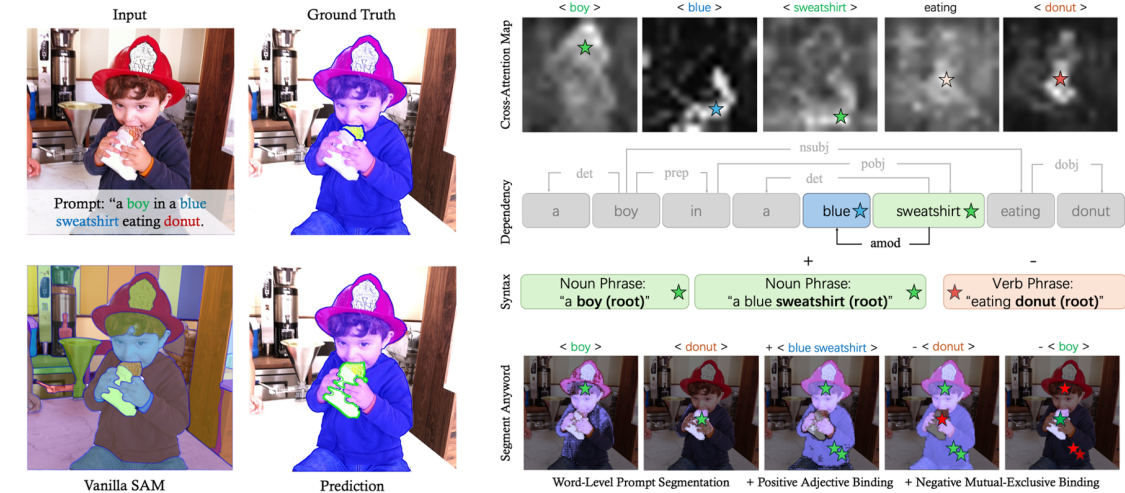
Segment Anyword: Framework



Plain Cross-Attention Cannot Segment Well



Linguistic Guided Visual Prompt Regularization



Quantitative Results

Table 2. Open-Set Language Grounded Segmentation Performance on GranDf Dataset validation and test split. “#Images” indicates the number of training images. “-” indicates the result is not reported from original paper.

Method	#Images	GranDf Val			GranDf Test		
		AP50	mIoU	Recall	AP50	mIoU	Recall
<i>Training End-to-End MLLM</i>							
1. BuboGPT	130M	19.1	54.0	29.4	17.3	54.1	27.0
2. Kosmos-2	9M	17.1	55.6	28.3	17.2	56.8	29.0
3. LISA	0.12M	25.2	62.0	36.3	24.8	61.7	35.5
4. GLaMM	11M	28.9	65.8	39.6	27.2	64.6	38.0
5. OMG-LLaVA	74K	26.9	64.6	-	26.1	62.8	-
<i>Fine Tuning MLLM</i>							
6. GLaMM _f	200K	30.8	66.3	41.8	29.2	65.6	40.8
7. OMG-LLaVA _f	200K	29.9	65.5	-	28.6	64.7	-
<i>Training-Free Prompt Learning</i>							
8. Segment Anyword	0	31.3	67.4	40.7	26.6	63.4	34.7
9. Segment Anyword _f w/ GT Text	500	30.2	65.9	42.4	31.1	64.1	33.9

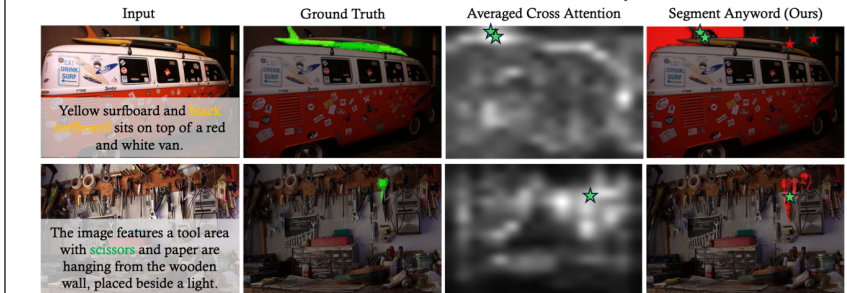
Predicate Segmentation

Predicate words without explicit semantic meaning, our method can be prompted to identify the predicate masks such as (Left) relation between object entities and (Right) human-object interaction.



Failure Case Analysis

Localization error due to limited cross-attention map resolution.



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